SIES GST | COMPUTER ENGINEERING | IDEA PRESENTATION 2026-27

MetroCheck AI

A first-pass screen for Legal Metrology declarations on a pack or a product listing. The model highlights seven facts. A written checklist scores them. A person confirms.

What this file is. Speaking notes for tomorrow. Each idea has a sentence you can say, then the technical line if a judge asks how. Every number below is either from this repository or from a named government reply. Lab scores are on 78 synthetic test documents. They are not live Amazon accuracy.

The 100 marks, in one line each

Rubric	20	Open with this
Innovation / ideation	20	The model does not decide guilt. It only finds spans. Rules score. A human signs.
Study design / methodology	20	Three readers, one held-out test of 78 texts. Regex-blind listings are forced into that test.
Feasibility	20	It runs on this laptop. No paid API. If a marketplace blocks the fetch, we stop.
Implementation and technology	20	Flask :5000, React :5173, SQLite, DistilBERT, seven JSON rules. Health shows hybrid.
Social relevance	20	Helpline Legal Metrology grievances and state inspection volumes are real. This is a reading aid for that desk.

Demo order if they say show it: health, then honey (ordinary labels, regex), then groundnut oil (origin missing), then Golden Grain atta (paraphrased labels, source = ner), then confirm, then PDF.

1. The questions, answered

1.1 What training means in this project

Training here means teaching an existing English model one narrow job: paint the words that hold a declaration. We did not teach the computer English from zero, and we did not train it to say compliant or guilty.

Say this. DistilBERT already knew English. We fine-tuned it to highlight seven facts on a pack: name, MRP, net quantity, who made it, country, made-on date, and use-before date. Each word gets a tag. The tag set has 15 labels.

If they want the mechanism. Base checkpoint distilbert-base-uncased, head DistilBertForTokenClassification. Seven entities times Beginning and Inside, plus Outside, equals 15 BIO tags (labels.py). Tokenizer is that checkpoint's WordPiece. Windows are 256 tokens with stride 64. Config in the saved model: hidden size 768. About 66 million parameters in the public base checkpoint. Inference is local. local_files_only. No hosted chat model.

1.2 Why this method

The legal words are stable. MRP, net quantity, manufactured by. A pattern search is the right tool for those words, because when it fires it is almost always right. Listings also rewrite the same fact. Pack price. Plant operator. COO. DOM. A pattern that only knows the textbook heading stays silent. That silence is the reason for the model.

Say this. On our 78 test documents the pattern search has precision 0.9291 and recall 0.2449. It filled 131 gold fields and missed 404. It made 10 false hits, all on product name. For MRP, quantity, manufacturer, origin, and dates its precision on that set is 1.0. Manufacturer recall for regex alone is 0.1282. The neural net's job is the miss, not a second opinion that overwrites a good hit.

If they want the mechanism. We refused to add Pack price and Plant operator as regex aliases. If those phrases were aliases, the demo could not show what the model adds. Hybrid merge (merge.py): agreement keeps the value and can keep the richer manufacturer string when one string contains the other and the shorter is at least 10 characters. Regex miss plus NER hit is accepted at medium confidence, capped at 0.78. Disagreement clears the value and sets status ambiguous. Both the regex path and the NER path run the same validators (validate_mrp, validate_country, validate_date, and the others). A span that is not a real amount, country, or date is dropped. The model cannot force a fake country through.

A chat model was the wrong tool for this job. It can write a country that the page never printed. This screen has to show the exact words it used, and the checklist has to stay a file a person can read. rules.json is ordinary if-then. Training does not edit it.

1.3 The real complaint record

This repository has no complaint file. The training texts are DemoMart packs and template paraphrases, not helpline dockets. The public record is large, and most of it is a different problem from the one this tool screens. Quote the source if a judge asks where a number came from. Two official replies differ by a few dockets. Say both. Do not average them.

National Consumer Helpline, all sectors, calendar year

Year	Grievances received	Source
2023	11,67,571	Lok Sabha reply, annex to AU1761
2024	13,43,412	Same reply. The reply says these dockets were disposed.
2025	17,71,622	Same reply.

E-commerce on that helpline

Year	E-commerce grievances	Note
2023	4,46,133	Lok Sabha reply on e-commerce grievances (AU2023).
2024	4,40,187	Same reply.
2025	5,11,196	Same reply.
2025, sector table	5,11,198	Rajya Sabha sector annex (AU1011). Two dockets apart from 5,11,196.

Those e-commerce numbers are mostly orders, delivery, and refunds. They are not labelling cases. This tool does not track a parcel and it does not obtain a refund. Say that before someone else does.

Legal Metrology, named as its own helpline sector, calendar 2025

Sector in the 2025 table	Received	What to do with it
E-commerce	5,11,198	Context. Mostly not our screen.
Food	91,012	Neighbour. Food law is a different statute.
Retail outlets	46,639	Shops, not only websites.
Legal Metrology	20,004	The helpline slice that names our statute. About 1.1% of 17.7 lakh.

Source: Rajya Sabha annex, sector-wise NCH complaints during 2025, Minister of State B. L. Verma.

Where the labelling cases actually pile up: inspections

Most weights-and-labels cases are booked by state officers on inspection, not by a person who already called 1915. Karnataka department data, reported by the Times of India on 25 December 2024:

Karnataka Legal Metrology	Cases booked
2021-22	26,431
2022-23	25,303
2023-24	16,806
2024-25, first seven months	9,660
2018-19 through November 2024	about 1.4 lakh cases
Inspections since 2018-19	4,67,948

Jammu and Kashmir's own 2024-25 progress report: 66,837 inspections, 204 consumer complaints to the department, 3,459 challans, 3,369 cases compounded, composition sum Rs 231.06 lakh. The gap between 66,837 inspections and 204 complaints is the point. The work is reading labels at inspection scale.

The Centre's e-commerce labelling file

A Lok Sabha reply states that from 2020 to 30 January 2026 the Department of Consumer Affairs issued 948 notices for e-commerce breaches of the Legal Metrology (Packaged Commodities) Rules, 2011, and recovered Rs 1.03 crore under Section 48 of the Act. A separate reply says 997 notices since 2020 for missing declarations such as country of origin, MRP, and net quantity, with the same Rs 1.03 crore figure. The count depends on the cut-off in the answer. The money figure is the one both answers repeat.

What this tool cannot see

A Lok Sabha reply for 2025-26 says states booked 381 cases of short delivery in packaged goods and recovered Rs 71.81 lakh, plus 2,046 cases of short selling of loose goods and Rs 1.47 crore. Reading a label cannot detect a light packet. Someone has to weigh it. Say that. It makes the scope honest.

1.4 How it helps, and who it is for

Say this. The law we screen against is the Legal Metrology (Packaged Commodities) Rules, 2011, under the Legal Metrology Act, 2009. A pre-pack is supposed to show MRP inclusive of taxes, net quantity, who manufactured or packed or imported it, the month and year, and country of origin where that declaration applies. The app looks for seven facts and shows the exact words. The score sorts the queue. It is not a conviction.

If they want the mechanism. Seven prototype rules in rules.json, version 1.0.0-prototype. The file itself says it is not a complete rulebook. Five required checks enter the score: REQ_MRP, REQ_NET_QTY, REQ_MANUFACTURER, REQ_COUNTRY, and REQ_MFG_OR_EXP (any of manufacturing date or expiry). CONF_LOW_MRP flags an MRP whose confidence is under 0.8. AMB_ANY flags every ambiguous field. Score = 100 times (passed checks / 5), then minus 8 for each high finding, 5 for medium, 2 for a review finding, clamped to 0-100. A missing required field both fails its check and adds a finding, so the score is stricter than a plain percentage. Overall status is needs_review, potential_issues, or clear_in_available_data.

Who	The job this screen helps	What they still do themselves
Legal Metrology staff	Read a pasted listing or a pack photo and see present, missing, or conflicting, with the quote attached.	Visit, seize, compound, prosecute. The PDF is a screening note.
Seller and catalog teams	Catch a listing that says pack price before it is live.	Fix the declaration. The Centre's notice file since 2020 is the public version of that failure.
Brand packaging associates	Check that a rewritten pack still carries the seven facts.	Own the artwork and the registration.
A consumer advisor or a college legal-aid cell	Show a shopper which line is missing before they call 1915.	Lodge the grievance. This app does not file it.

A sentence that costs marks. Saying we solve consumer complaints. The 2025 helpline logged about 5.11 lakh e-commerce grievances and 20,004 under Legal Metrology. We sit in the smaller queue, and only on the part that is visible in text. Replacement sentence: this is a reading aid for declaration completeness, for the officer and the seller who have to look at many labels.

2. Architecture decisions

Two programs. The brain is Python. The face is the web page. They talk only through /api. Everything heavy sits on this machine. There is no cloud account in the scan path.

Piece	Where	Why this, in plain words	Why this, technically
API	Flask, 127.0.0.1:5000	One program can load the model and answer the page.	Same process as PyTorch. run.py keeps debug off so the model is not loaded twice. NER warmup is about a minute.
Screen	React + Vite, 127.0.0.1:5173	Buttons, the scan form, the result.	Hash routes: landing, #scanner, #rules, #history. Vite proxies /api to port 5000. Port 5174 is only a fallback if 5173 is taken.
Memory	SQLite	One file. No database server to install on demo day.	Tables: products, extracted_fields, findings, reports. DATA_DIR is D:\legal-metrology\data.
Reader	Regex, always	The strict highlighter. It does not invent.	patterns.py plus validators in extractors.py. Bare 199 is not MRP. MRP 1169 is not captured as 116.
Second reader	DistilBERT NER	Reads the same fact in other words.	Weights in backend/app/ml/ner_model, gitignored. EXTRACTOR_MODE=hybrid. If weights are missing, the code falls back to regex and warns. It does not download.
Judge of the checklist	rules.json	A marking scheme a person can open.	evaluate_rules in rules.py. Four rule types: presence, any_of, confidence, ambiguous.

Piece	Where	Why this, in plain words	Why this, technically
Practice shop	DemoMart, /demo/	The demo does not depend on Amazon being polite.	Local HTML. Honey uses textbook labels. Atta, pepper, ghee and the other regex-blind SKUs use paraphrases.
Photo text	Tesseract, optional	A photo of a pack, when the program is installed.	On this machine ocr_tesseract_available is false. Demo path is URL and pasted text.

2.1 The path of one scan

1. The page sends POST /api/scan/url, or /text, or /image, or /sample.
2. pipeline.py gets words. A DemoMart or ordinary page uses the scraper. A blocked Amazon, Flipkart, or Zepto page stops. Pasted text is already words. An image would go through OCR.
3. Regex runs. If mode is hybrid and the weights load, NER runs. merge_regex_ner combines them. Page text and OCR, when both exist, are merged by merge_fields with the same idea: agree, keep one, or mark ambiguous.
4. rules.py writes findings and a score.
5. storage.py saves the product, the fields, and the findings.
6. The screen shows evidence. A person confirms or rejects. A PDF is generated only when they ask.

Say this. Evidence is the exact snippet, not a vibe. If two sources disagree, the value is blank and the status is ambiguous. The person sees both candidates.

If they want the mechanism. Evidence is an object: matched_text, source, start, end, pattern_id. The React screen must print matched_text. Printing the object itself crashes the page. That bug was fixed in ProductDetail.jsx. API responses carry a disclaimer: findings are potential issues, and a human reviewer must confirm before any enforcement.

2.2 Decisions a judge may challenge

Why stop when Amazon shows a robot check

Say this. Their site is allowed to refuse a robot. We stop in about a second and tell the person to paste the text or open DemoMart. We do not solve the puzzle, and we do not hide behind a proxy.

If they want the mechanism. scraper.py treats a bot wall as error_code marketplace_blocked. Hard HTTP blocks skip Playwright so the screen does not hang for tens of seconds. The UI has a blocked panel with a DemoMart link and a switch to paste text. A test posts robot-check HTML to /api/scan/url and expects ok false, that error code, and a DemoMart fallback.

Why the model is not allowed to overrule a clear regex hit

Say this. If the pattern already found MRP Rs. 349, a second guess must not silently replace it. If the second guess differs, a person looks. That is safer than a smoother score.

If they want the mechanism. On the 78-document test, NER-only F1 is 0.9925 and hybrid F1 is 0.9839. Hybrid gives up a little recall so a conflict stays visible. Precision of both on that set is 1.0. Quote the field score, not only the token score. Token seqeval on the test set is precision 0.9943, recall 0.9924, F1 0.9934. A field counts as a hit only when status is detected and the normalized value matches gold.

Why SQLite and a laptop, not a platform

The user of a college demo is one reviewer in the room. A single file database and a local model match that user. The D: drive holds the virtual environment, Playwright's browser, the database, and the Hugging Face cache, so the project disk stays the source code. Health is the proof the right Python is running: ok true, ner_available true, extractor_mode hybrid. System Python without the weights still serves the site and turns the model off. That is visible, not silent.

3. How the model is trained

3.1 The dataset

make_dataset.py builds the files. Sources are regex-blind gold records, sample packs, and synthetic paraphrases (default 480 synthetic documents, clamped between 300 and 600). Rows are deduplicated by id. The split is by whole document, 70 / 15 / 15, seed 42. Regex-blind SKUs always go to test, so the exam includes phrases the pattern cannot see.

Split	Documents	Role
Train	343	The model sees these.
Validation	73	Watched during training.
Test	78	The published comparison. Includes the regex-blind listings.
regex_blind.jsonl	4 seed texts	The paraphrased packs written so regex should miss. Held into test.

Total documents in the three splits: 494. Gold is templated DemoMart text plus paraphrases. It is not a scraped corpus of Indian packs. Say that in the same breath as the F1.

3.2 What one training example looks like

Say this. We mark the characters of each fact in the text. The trainer cuts the sentence into word-pieces and copies those marks onto the pieces. The model then practises predicting the mark for every piece.

If they want the mechanism. Character spans become BIO tags. Hugging Face Trainer. Script defaults, from train_ner.py: 4 epochs, batch size 8, learning rate 5e-5, max length 256, stride 64, seed 42. CPU if there is no CUDA. The saved folder has config and weights. It has no trainer_state.json, so these are the script defaults, not a recovered log of the exact run. Optional flag --multilingual switches the base to bert-base-multilingual-cased. The shipped model is the DistilBERT one.

3.3 The score you are allowed to quote

System	Precision	Recall	F1	What it means on these 78 texts
Regex only	0.9291	0.2449	0.3876	Careful, and silent on paraphrases. 131 true fields, 10 false, 404 missed.
NER only	1.0000	0.9850	0.9925	Finds the paraphrases. Still a lab set.
Hybrid	1.0000	0.9682	0.9839	The shipped mode. Slightly lower recall, conflicts stay empty.

Field, hybrid	Precision	Recall	F1	Support
product_name	1.00	0.8590	0.9241	78
mrp	1.00	0.9872	0.9935	78
net_quantity	1.00	0.9872	0.9935	78
manufacturer	1.00	0.9872	0.9935	78
country_of_origin	1.00	0.9865	0.9932	74
manufacturing_date	1.00	0.9868	0.9934	76
expiry_date	1.00	0.9863	0.9931	73

Product name is the weak field. Names are free text. MRP and quantity are tighter. Regex recall on manufacturer / origin / manufacturing date / expiry is 0.1282 / 0.2703 / 0.2632 / 0.2740. Hybrid recall on those four is 0.9872 / 0.9865 / 0.9868 / 0.9863. That gap is the methodology result.

A sentence that costs marks. Saying 99 percent accuracy on Amazon, or that the AI checks compliance. Replacement: on 78 practice texts we wrote, hybrid field-level F1 is 0.9839. The model finds spans. The checklist decides what is missing. A person confirms. We have not measured live marketplace pages, and the scraper is designed to stop when those sites block it.

3.4 The merge, as a table you can draw

Regex	NER	What the product does
Already ambiguous	Anything	Stay ambiguous. A person looks.
Same value	Same value	Keep it. Confidence can rise by 0.04, capped at 0.99. Source merged.
Hit	Miss	Keep the regex value.
Miss	Hit	Keep the NER value. Confidence capped at 0.78. Reason: regex miss, accepted NER span.
Different values	Different values	Value cleared. Status ambiguous. Both candidates stored.
Miss	Miss	not_detected. Nothing is invented.

Same value includes a manufacturer name contained in a longer address, when the shorter string is at least 10 characters. The richer string is kept.

4. Words you can explain both ways

Word	Say this	If they want the mechanism
Regex	A search for an exact shape of text, such as MRP and then an amount.	patterns.py. Precision-first. Empty stays empty.
Token	A piece of a word the model reads.	WordPiece. Himalayan can split. The tokenizer ships with DistilBERT. We did not train a second tokenizer.
NER	Named entity recognition. Finding which span is the price, which span is the country.	Token classification. Not a chatbot. Not a verdict.
BIO	Beginning, Inside, Outside. A tag for each piece.	B-mrp starts the price. I-mrp continues it. O is everything else. 15 tags.
Fine-tuning	We started from a model that knew English and taught it our 15 tags.	494 short documents cannot teach English. They can teach a label set.
Precision	When it speaks, how often it is right.	True fields divided by the fields it filled. Regex 0.9291 on the test set.
Recall	Of the facts that were really there, how many it found.	Regex 0.2449. Hybrid 0.9682. Same 78 documents.
F1	One number that punishes both false hits and misses.	Harmonic mean of precision and recall. Quote field F1, and say the set.
Hybrid	Pattern first. Model only fills a hole. A fight goes to a person.	merge_regex_ner. Default EXTRACTOR_MODE.
Ambiguous	Two readers disagreed, so we refused to pick.	value null, alternatives kept, rule AMB_ANY raises needs_review.
Rule engine	The marking scheme. If MRP is missing, that is a finding.	JSON. presence, any_of, confidence, ambiguous. Training never writes this file.
Confidence	How hard the reader is willing to lean on that hit.	NER-only hits are capped at 0.78 so they stay a medium hit. Low MRP under 0.8 raises CONF_LOW_MRP.
OCR	Reading letters off a photograph.	Tesseract. Not installed here. The demo does not depend on it.
SQLite	The notebook of past scans, stored as one file.	Four tables. No separate database process.
Proxy	The web page asks /api, and Vite forwards that to Flask.	vite.config.js, target 127.0.0.1:5000.
Fail closed	If we are not allowed to read the page, we stop and say so.	marketplace_blocked. No CAPTCHA solver, no proxy farm, no cookie replay.
Validator	A second gate that throws away a span that is not a real price, date, or country.	Shared by regex and NER. The model proposes. The validator can refuse.
Score	A queue priority from 0 to 100.	Passed required checks over 5, then small penalties. Not a legal certificate. Status clear_in_available_data still means in the text we saw.

5. How to take each 20

Speak in this order inside each rubric: the decision, the evidence on the laptop, the limit. Judges reward a team that knows the edge of the work.

5.1 Innovation and ideation (20)

Say this. The idea is the split of jobs. A small model reads messy words. A file of rules applies the checklist. A person signs. Most student AI projects let the model do all three. That is where invented facts come from. We kept them apart.

If they want the mechanism. Show one regex-blind SKU and point at source = ner, then show a conflict path if you have one, then open rules.json and show that training does not touch it. The second idea is fail-closed scraping. Stopping at a marketplace wall is a product decision. It keeps the project on the right side of the site's rules and it keeps the demo honest.

A sentence that costs marks. We used AI to check legal compliance, or we automated enforcement. Replacement: we automated the reading of seven declarations, and we left the legal judgment with a person and a written rule.

5.2 Study design and methodology (20)

Say this. We wrote the practice texts, then we hid a test. The hardest listings, the ones whose wording the pattern cannot see, were forced into the test. We scored three systems on that same test: pattern only, model only, and both. A hit counts only when the cleaned-up value matches.

If they want the mechanism. Document-level split, seed 42, so a sentence from one pack does not land in both train and test. Field-level precision, recall, and F1, plus a token seqeval number that we treat as secondary. The weak cell is product-name recall 0.8590. Mention it. External validity is the limit: the test is paraphrases of our templates. A live catalog can look different. We did not claim that measurement.

If they ask why F1 is so high: the model was tested on text written in the same style as the training text, and the label set has only seven fields. High F1 on that set shows the reader learned our paraphrases. It does not show performance on the Indian internet. The useful comparison is the gap against regex on manufacturer and dates, on that same set.

5.3 Feasibility (20)

Say this. You can watch it on this laptop in two minutes. No key, no bill, no account. If the marketplace refuses, the fallback is pasted text or DemoMart, and both are in the product, not in a slide.

If they want the mechanism. START-EVERYTHING.ps1, or the two scripts. Health URL. D: venv so torch is present. Known gap, say it first: Tesseract is not installed, so photo OCR is off. Playwright exists for pages that allow a browser, and it is skipped for localhost and for hard blocks so the demo cannot hang. The machine is a student laptop. Training defaults are CPU-capable. Inference of one page does not need a GPU.

A sentence that costs marks. We will scale to all of Amazon next week. Replacement: the feasible product today is a reviewer's desk for text we are allowed to read. Marketplace automation stops at the wall, on purpose.

5.4 Implementation and technology (20)

Name files, not slogans. A technical judge is listening for whether you can walk the path.

If they ask	Walk here
Where does a scan enter?	backend/app/routes/api.py then services/pipeline.py
Where are the patterns?	field_extractor/patterns.py and extractors.py
Where is the model called?	field_extractor/ner_extractor.py, local_files_only
Where is the policy?	field_extractor/merge.py, function merge_regex_ner
Where is the law-shaped checklist?	backend/data/rules.json and services/rules.py
Where is memory?	services/storage.py, four tables
Where is the screen?	frontend/src, hash routes, Vite proxy
Where is the practice shop?	backend/app/demo_store/
How do I know the model is on?	GET /api/health, ner_available and extractor_mode
What did the tests lock?	Marketplace block returns marketplace_blocked. MRP must not turn 1169 into 116.

Stack in one breath: Python, Flask, dataclasses for each extracted field, React 18, Vite 5, SQLite, PyTorch and Hugging Face Transformers for the token classifier, ReportLab for the PDF, optional Tesseract, optional Playwright. No OpenAI, no Gemini, no paid inference.

A sentence that costs marks. The model is a large language model that understands the Legal Metrology Act. Replacement: it is a distilled BERT with a 15-way tag on each word-piece. The Act is not inside the weights. The checklist is a JSON file we wrote, and it is a prototype, not the full Rules.

5.5 Social relevance (20)

Say this. A shopper is supposed to see the price ceiling, the quantity, who made it, and where it came from, before they pay. Officers already book these cases in the tens of thousands in a single state. The helpline's own Legal Metrology line was 20,004 grievances in 2025. The tool helps the person who has to read the next label, and it helps a seller fix the line before a notice.

If they want the mechanism. Use three numbers and one limit. Karnataka: 16,806 cases in 2023-24, and 4,67,948 inspections since 2018-19. J&K 2024-25: 66,837 inspections against 204 department complaints. Centre, e-commerce labelling notices since 2020: 948 as on 30 January 2026, or 997 in a later reply, Rs 1.03 crore compounded. Limit: short weight (381 packaged cases, Rs 71.81 lakh, in the 2025-26 reply) needs a weighing. We do not pretend to do that. Second limit: the big helpline number is e-commerce refunds. We do not claim it.

Who we are speaking to, in the order that matches the evidence: the inspector's office, the seller's catalog desk, the brand's packaging desk, and a student or advisor helping someone call 1915. The college build is a working desk for that reading job, sized to one laptop.

6. Ninety seconds, then the demo

Say this, almost as written.

When you buy a packed product in India, the pack has to show the maximum price, how much is inside, who made it, and where it came from. On a website those lines are often missing or written in other words. A person cannot open thousands of pages by hand.

MetroCheck reads the page, or text you paste, and highlights seven facts. A pattern search catches the official wording. A small trained model catches the rewritten wording. If they disagree, we leave the box empty. A checklist of seven prototype rules then scores what is missing. You confirm. Then you can download a PDF. The model does not decide that anyone broke the law.

On 78 practice texts, the pattern search found about a quarter of the fields and was rarely wrong. The combined reader reached a field F1 of 0.98 on that same practice set. That number is not live Amazon.

The need is already in the public record. The national helpline tagged 20,004 grievances as Legal Metrology in 2025, inside 17.7 lakh complaints of every kind. Karnataka booked 16,806 weights and labelling cases in 2023-24. We built the first screen for the people who read those labels.

Demo clicks

Step	What they should see	The sentence
Open /api/health	ner_available true, extractor_mode hybrid	The right Python is loaded. The model is on.
Scan hive-organic-honey-500g	High score. Labels like MRP and Manufactured by.	Textbook wording. The pattern search is enough. It does not invent.
Scan pure-groundnut-oil-1l	Country of origin missing	A missing declaration is a finding, not a guess.
Scan golden-grain-atta-5kg	Fields filled, source ner. Wording like stated consumer price, plant operator.	The pattern is silent. The model paints the span. Confidence stays medium.
Confirm, then PDF	Reviewer action stored. Report downloads.	A human signs. The PDF is the note, not the judgment.

If a live Amazon URL is demanded: run it, let it fail closed, and read the blocked panel. That failure is the design. Then paste a paragraph or return to DemoMart.

Five questions, short answers

Question	Answer
Why not ChatGPT?	A chat model can write a country the page never had. We need a span, a validator, and a rule file. All three run offline.
Why is the score almost perfect?	Because the test is our own paraphrases, 78 documents, seven fields. The honest result is the gap versus regex, not a claim about Amazon.
Why not add more patterns?	We could. We did not, for the phrases that define the experiment. Adding them would hide the model and would still miss the next paraphrase.
Can this fine a seller?	No. Status values are potential_issue and needs_review. The disclaimer is on the API and on the report. Section 48 and compounding sit with the officer.
What is unfinished?	Photo OCR on this machine. A full encoding of the Rules, not seven prototype checks. A test set of real packs we did not write. Live marketplace fetch, which we refuse to force.

7. Sources for the public numbers

Project numbers are from this repository: backend/app/ml/RESULTS.md, metrics.json, make_dataset.py, train_ner.py, merge.py, rules.json, rules.py, storage.py. Train 343, validation 73, test 78 were counted from the JSONL files on this machine.

Public record, in the words of the replies:

Lok Sabha, annex to AU1761. NCH grievances received: 2023 11,67,571; 2024 13,43,412; 2025 17,71,622. The reply says the dockets were disposed. sansad.in, file AU1761_u7HUSE.pdf.

Lok Sabha, annex to AU2023. E-commerce grievances: 2023 4,46,133; 2024 4,40,187; 2025 5,11,196. Same reply: 948 notices from 2020 to 30 January 2026 for e-commerce breaches of the Packaged Commodities Rules, compounding Rs 1.03 crore under Section 48. File AU2023_v5xE5g.pdf.

Rajya Sabha, annex to AU1011. Sector-wise NCH during 2025: E-commerce 5,11,198; Food 91,012; Retail outlets 46,639; Legal Metrology 20,004. File AU1011_8wOSPV.pdf.

Lok Sabha, annex to AS361. From 2020 till the date of that reply, 997 notices for non-declaration of mandatory information such as country of origin, MRP, and net quantity, compounding Rs 1.03 crore. File AS361_5fKGRD.pdf. Different cut-off from the 948 figure. Same money figure.

Times of India, 25 December 2024, Bengaluru. Karnataka Legal Metrology department data: cases 26,431 (2021-22), 25,303 (2022-23), 16,806 (2023-24), 9,660 in the first seven months of 2024-25; about 1.4 lakh cases from 2018-19 to November 2024; 4,67,948 inspections since 2018-19.

Legal Metrology Department, Jammu and Kashmir. Progress report 2024-25: 66,837 inspections, 204 consumer complaints, 3,459 challans, 3,369 compounded, Rs 231.06 lakh composition. legalmetrology.jk.gov.in, enforcement data PDF updated May 2025.

Lok Sabha, annex to AU1636. 2025-26: 381 cases of short delivery of packaged commodities, Rs 71.81 lakh compounded; 2,046 cases of short selling in loose form, Rs 1.47 crore. The same reply describes state enforcement of MRP, net quantity, manufacturer details, month and year, and consumer-care declarations.

Statute named, not reproduced: Legal Metrology Act, 2009, and Legal Metrology (Packaged Commodities) Rules, 2011. The project's rules.json is a prototype subset and says so in its disclaimer.

Last line if the room goes quiet. It is a screening desk. The pattern is strict. The model only reads other words for the same fact. The checklist is a file. The person signs. The 0.98 is on practice text we wrote. The inspections and the 20,004 helpline grievances are why the desk is worth building.